12.9 Gauss公式与Stokes公式 (空间) A[P, Q, R].

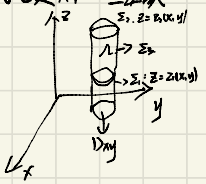
Gauss公式：设空间闭区域Λ由分片光滑的闭曲面Σ所围成，Σ方向取外侧，函数P, Q, R在Σ上有连续的一阶偏导数，则有

$$\oiint_{\Sigma} P\,dydz+Q\,dzdx+R\,dxdy = \iiint_{\Omega}\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dx\,dy\,dz.$$

$$\Longleftrightarrow \oiint_{\Sigma} \vec{A}\cdot d\vec{S} = \iiint_{\Omega} \nabla\cdot\vec{A}\,dv.$$

Case1

Pf：Λ是XY-型区域　Σ = Σ1 + Σ2 + Σ3　只证 $\oiint_{\Sigma} R\,dxdy = \iiint_{\Omega}\frac{\partial R}{\partial z}\,dx\,dy\,dz$.



Λ: $z_1(x,y) \le z \le z_2(x,y)$, $(x,y)\in D_{xy}$.

Σ3: $H(x,y)=0$　$\cos\gamma = 0$

Σ1: $z = z_1(x,y)$ 下侧

Σ2: $z = z_2(x,y)$ 上侧

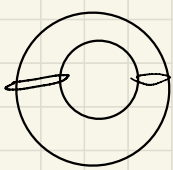
$$\iiint_{\Omega}\frac{\partial R}{\partial z}\,dx\,dy\,dz = \iint_{D_{xy}} dx\,dy \int_{z_1(x,y)}^{z_2(x,y)}\frac{\partial R(x,y,z)}{\partial z}\,dz$$

$$= \iint_{D_{xy}} dx\,dy\,\big(R(x,y,z_2(x,y)) - R(x,y,z_1(x,y))\big).$$

$$\oint_{\Sigma} R\,dx\,dy = \iint_{\Sigma_1} R\,dx\,dy + \iint_{\Sigma_2} R\,dx\,dy + \iint_{\Sigma_3} R\,dx\,dy$$

$$= \iint_{D_{xy}} R(x,y,z_2(x,y))(-1)\,dS + \iint_{D_{xy}} R(x,y,z_2(x,y))\,dS + 0$$

$$= \iint_{D_{xy}} \big(R(x,y,z_2(x,y)) - R(x,y,z_1(x,y))\big)\,dx\,dy.$$



Case 2. 若Λ不是XY-型区域

将闭域分为上下两部分，在截面处，法向量方向正好相反，相互抵消，则Gauss公式依然成立

Remark. 若令P = x, Q = y, R = z 由Gauss公式可得 $\oiint_{\Sigma} x\,dydz + y\,dzdx + z\,dxdy = 3\iiint_{x^2+y^2+z^2\le R^2} dx\,dy\,dz = \frac{4}{3}\pi R^3$

例1. 计算 $\oiint_{\Sigma} x^3\,dydz + y^3\,dzdx + z^3\,dxdy$　Σ: $x^2+y^2+z^2 = R^2$

原式 $= 3\iiint_{\Omega}(x^2+y^2+z^2)\,dx\,dy\,dz$.

$= 3\int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^R \rho^2\sin\varphi\,d\rho$.

$= \frac{3}{5}R^5\int_0^{2\pi} d\theta \int_0^{\pi}\sin\varphi\,d\varphi$

$= \frac{12\pi}{5}R^5$



变式1. 求 $I = \oiint_{\Sigma}\frac{x^3\,dydz + y^3\,dzdx + z^3\,dxdy}{\sqrt{x^2+y^2+z^2}}$，Σ是球面 $x^2+y^2+z^2 = R^2$ $(R>0)$ 外侧

原式 $= \frac{1}{R}\oiint_{\Sigma} x^3\,dydz + y^3\,dzdx + z^3\,dxdy = \frac{12}{5}\pi R^4$

变式2. 求 $I=\oint_{\Sigma} x^3dydz+y^2dzdx+z\,dxdy$

$=\iiint_{\Omega}(3x^2+2y+1)\,dxdydz$

$=\iiint_{\Omega}3x^2\,dxdydz+\iiint_{\Omega}dxdydz$

$=\iiint_{\Omega}(x^2+y^2+z^2)\,dxdydz+\iiint_{\Omega}dxdydz$

$=\frac{4\pi R^5}{5}+\frac{2\pi R^3}{3}$

例2. 求 $I=\oint_{\Sigma} x^3dydz+y^3dzdx+z^3dxdy$，$\Sigma$ 是正方体 V. $0\le x\le a$ $0\le y\le a$. $0\le z\le a$. 的全表面外侧

$=\iiint_{V}(2x+2y+2z)\,dxdydz$

$=2\iiint_{V}(x+y+z)\,dxdydz$

$=2\int_0^a dx\int_0^a dy\int_0^a(x+y+z)\,dz$ $=2(\bar{x}+\bar{y}+\bar{z})\iiint_{V}dxdydz$

$=3a^4$

变式1 Σ 是球面 $x^2+y^2+z^2=R^2$，$(R>0)$ 外侧

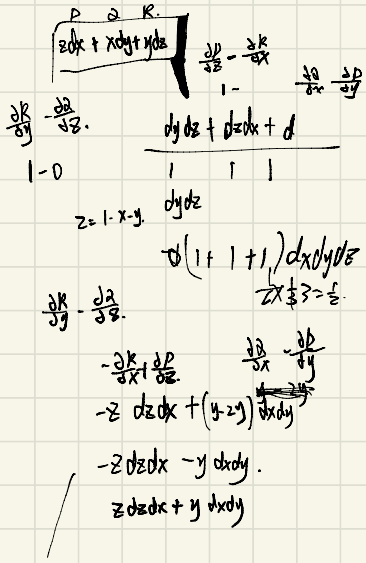
$I=2\iiint_{\Omega}(x+y+z)\,dxdydz=0$

变式2. Σ 是锥面 $z=\sqrt{x^2+y^2}$ 与平面 $z=1$ 的围立体的 表面外侧

$I=2\iiint_{\Omega}(x+y+z)\,dxdydz$

$=2\iiint_{\Omega}z\,dxdydz$

Stokes公式：若P, Q, K在分片光滑曲面Σ（包括边界分段光滑曲线L）上连续且具有连续的偏导数，则



1.

设 $(\vec{a}\times\vec{b})\cdot\vec{c}=2$，则 $[\vec{a}+\vec{b}]\times[\vec{b}+\vec{c}]\cdot[\vec{c}+\vec{a}]=$

$$\begin{vmatrix} x_a+x_b & y_a+y_b & z_a+z_b \\ x_b+x_c & y_b+y_c & z_b+z_c \\ x_c+x_a & y_c+y_a & z_c+z_a \end{vmatrix} \begin{matrix} ① \\ ② \\ ③ \end{matrix} = \begin{vmatrix} ① & x_a+x_b \\ ② & x_b+x_c \\ ②-① & x_c-x_b \end{vmatrix} = \begin{vmatrix} ① & x_a+x_b \\ ② & x_b+x_c \\ ③-①+③ & 2x_c \end{vmatrix} = \begin{vmatrix} x_a & \\ x_b & \\ 2x_c & \end{vmatrix} = 4.$$

2. 分别求母线平行于 X 轴及 Y 轴而且通过曲线 $\begin{cases} 2x^2+y^2+z^2=16 \\ x^2+z^2-y^2=0 \end{cases}$ 的柱面方程

解：消去 X，得 $3y^2-z^2=16$ 即是所求母线平行于 X 轴且过已知曲线的柱面方程

$\int_0^{2\pi} d\theta \int_0^{a} \left(r\cos\theta + \frac{2r\sin\theta(1+r\cos\theta)}{\sqrt{R^2-r^2}} \right) r\,dr$

$r^2\cos\theta + \frac{2r^2\sin\theta}{\sqrt{R^2-r^2}} + \frac{2r^3\sin\theta\cos\theta}{\sqrt{R^2-r^2}}\,dr$

$2r\sin\theta\cos\theta$

$\frac{a^3}{3}\cos\theta +$

$\frac{\sqrt{3}}{3}$

$\frac{x^2+y^2+x^2-x^2-y^2}{z}$

$\frac{R}{z}(xz+2xy+2yz)\cdot\frac{-x}{z}$

$\left(x+\frac{2y(1+x)}{\sqrt{R^2-x^2-y^2}}\right)dxdy$

$(xz+2xy+2yz)\sqrt{2x^2+2y^2+1}\cdot$

$\left|(xz+2xy+2yz)\,dS\right.$

$(z\cdot(-2x)+2x(-2y)+2y)$

$(x-\frac{a}{2})^2+y^2=(\frac{a}{2})^2$

$x^2-ax+y^2=0$

$x^2+y^2=ax$

$\frac{zx+2xy+2y}{\sqrt{R^2-x^2-y^2}}$

$z-x+2x$

$z=\sqrt{a^2-x^2-y^2}$

$\iint_\Sigma u\left(\frac{\partial v}{\partial x}\cos\alpha + \frac{\partial v}{\partial y}\cos\beta + \frac{\partial v}{\partial z}\cos\gamma\right)dS$

$u\left(\quad\right)'+u'\left(\quad\right)$

$\frac{\partial^2 v}{\partial x^2}$

$\int f(x)dydz$

$\int f(x)\,ds$

$g(u)ds$

$\int f(x)$

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